

Infosys Hackathon

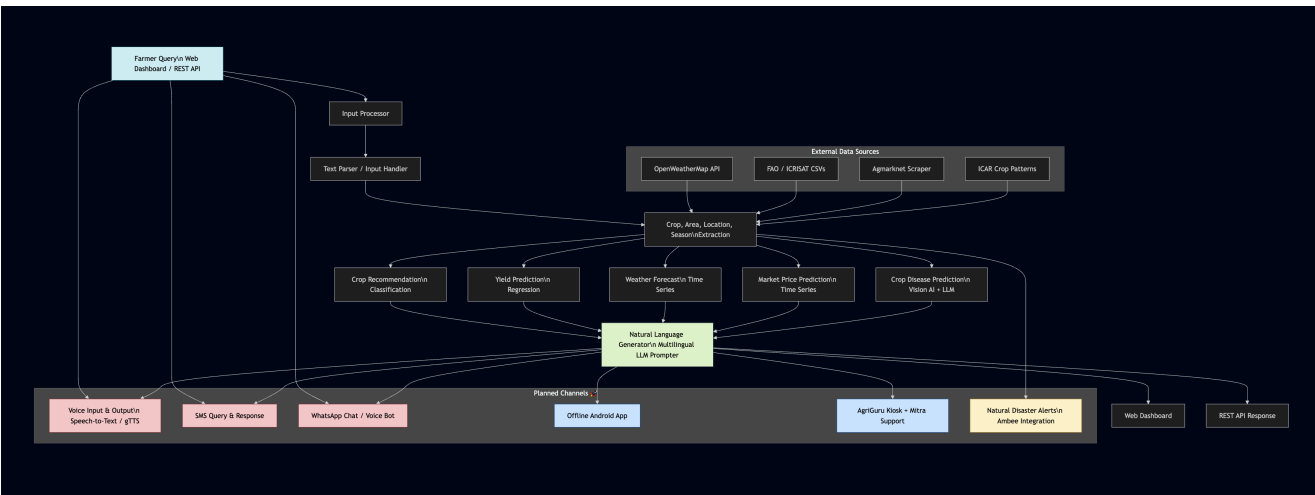
Purpose & Vision

To Give every smallholder farmer in the World an on-demand, AI-generated advisory that answers four questions:

1. *What crop should I plant now?*
2. *How much yield can I expect?*
3. *What weather should I prepare for?*
4. *What price can I likely sell at?*

The platform delivers these insights in local languages via simple web / SMS / voice channels, but the MVP focuses on a **REST API + React dashboard** that demo end-to-end intelligence.

Process Flow :



AI Component	Function	Model Type
Crop Recommendation Engine	Recommend optimal crop per location & season	Classification
Yield Prediction	Estimate yield based on crop, soil, and weather	Regression
Weather Forecast Analysis	Forecast rainfall/temperature trends	Time Series
Market Price Forecasting	Predict local mandi prices	Time Series
Natural Language Generator	Generate human-like advisory (in all Languages)	LLM Prompt

AI Component	Function	Model Type
Crop Disease Prediction	Predict any disease that the crop is prone to and suggest	Vision AI + LLM Analysis

Datasets we are using :

Source	Use	Link
OpenWeatherMap	Real-time weather	API
FAO Crop Data	Crop-soil-weather-yield	CSV
ICRISAT / ICAR	Indian crop patterns	CSV
Agmarknet	Price data	CSV / scrape
Ambee	Forecast	API

Problem: Farmers may lack smartphones, literacy, or digital fluency.

Future Scope : (E2E)

✅ Multi-Channel Access Strategy (Tech + UX + Human)

🟢 1. Voice-Enabled Advisory (IVR or WhatsApp Audio)

- Farmers call a local number or send a voice message.
- Backend uses:
 - Speech-to-Text : Google Speech API or `whisper.cpp` (open-source)
 - Process query using AI/crop logic
 - Reply with:
 - **Text** if farmer can read
 - **Voice** using `gTTS` or Google TTS

✅ Simple, no app needed. Works on basic phones.

🟢 2. SMS-Based Interface

- Farmer sends SMS:


```
ADVISORY Akola Cotton 2 acre
```
- You parse and reply:
 - ```
Rain expected in 3 days. Delay sowing. Use Urea: 2kg/acre.
```

✅ Works on basic phones, very reliable.

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### ● 3. WhatsApp Chatbot (or Voice Bot)

- Use Twilio + LangChain or Meta's WhatsApp API
- Allow input via:
  - Buttons (yes/no)
  - Audio messages
- Use `whisper` to convert voice to text and respond.

✓ Familiar interface for younger farmers.

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### ● 4. Offline Android App (with periodic sync)

- Lightweight Flutter/React Native app with:
  - Minimal UI: icons, images
  - Voice prompts
  - Local language
- Syncs data only when online

✓ For areas with poor connectivity

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### ● 5. Kiosk/Mitra Model (Hybrid Human Support)

- Place a **tablet + solar-powered kiosk** in the panchayat/mandi
- Trained local youth ("AgriGuru Mitra") helps farmers use app

Planning to integrate Natural Disasters Prediction, Wildfire Prediction using Ambee (In Progress)

Ambee API Key :

4733e8c05c4a5561d69413deb345f3f7c293ca72f1d7fa559392be79835afe34

### Out of Scope as of Now :

1. We don't have real sensors or IoT Devices, hence relying on the Data sources which are existing and taking data from authentic data sources
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Predict HQ

<https://github.com/nanditha213/Natural-Disaster-Prediction>

<https://chatgpt.com/c/68753e35-38fc-8002-804a-57c0efbc8e76>